MTH 331(A) Dr. James Ashe
Final Exam Review Final Exam will be be given on January 15^{th} 5pm-7pm, 2012

Reading and writing sigma notation

1. Write the infinite series, $0.3 + 0.03 + 0.003 + \cdots$, using sigma notation (i.e. using Σ) .

2. Write the first 3 terms of the sequence of partial sums of the series $\displaystyle\sum_{k=1}^{\infty} \frac{3^k}{2}$.

Convergence/Divergence of series

Evaluate the geometric series or state that it diverges.

3. $\displaystyle\sum_{k=0}^{\infty} \frac{2^k}{7^k}$

4. $\displaystyle\sum_{m=2}^{\infty} \left(\frac{1}{3}\right)^m$

Determine the radius of convergence of the following power series.

5. $\displaystyle\sum \frac{x^k}{2^k}$

Using the Squeeze Theorem, find the limit of the following sequence or determine that the limit does not exist.

6. $\displaystyle\lim_{n\to\infty} \left\{\frac{\sin^2 n}{2^n}\right\}$

Use the Divergence Test to determine whether the following series diverge, or state that the test is inconclusive.

7. $\displaystyle\sum_{k=1}^{\infty} \frac{k^4 - 1}{k^4 + 10}$

Use the Ratio Test to determine whether the following series converges.

8. $\displaystyle\sum_{k=1}^{\infty} \frac{3k + 3}{6(k + 1)}$

Taylor polynomials

9. Let $f(x) = 2\cos(x)$. Find the Taylor polynomial of order $n = 3$ for $f(x)$ centered at 0.
- **a.** Use the Taylor polynomial from above to approximate $2\cos(0.1)$.
- **b.** Compute the absolute error in the approximation assuming the exact value is given by a calculator.

Parametric functions

Determine dy/dx in terms of t and evaluate it at the given value of t.

10. $x = 2t$, $y = 4t^2$; $t = 1$.

Give a set of parametric equations that describes the curve.

11. A circle centered at $(0, 0)$ with radius 6, generated clockwise.

Calculus in polar coordinates

Locate the following polar coordinates on a graph.

12. A:$\left(2, \frac{\pi}{4}\right)$, B:$\left(-2, \frac{\pi}{4}\right)$, C:$\left(2, \frac{\pi}{4} + \pi\right)$, D:$\left(2, \frac{\pi}{4} + 2\pi\right)$

*Express the following **polar** coordinates in **Cartesian** coordinates.*

13. $\left(-4, \frac{2\pi}{3}\right)$

*Express the following **Cartesian** coordinates in **polar** coordinates in at least **two** different ways.*

14. $(-1, 1)$

Find the slope of the line tangent to the following polar curves at the given point using calculus.

15. $r = 2\cos 2\theta$; $\left(2, \dfrac{\pi}{2}\right)$.

Vectors

Let $\mathbf{u} = \langle 2, 4, 1\rangle$ *and* $\mathbf{v} = \langle 1, -3, 2\rangle$. *Perform the indicated operations.*

16. $\mathbf{u} - 2\mathbf{v}$

17. $\mathbf{u}\cdot\mathbf{v}$

18. $\mathbf{u} \times \mathbf{v}$

Use the graph to draw the following vectors (provide labels). Make certain to indicate direction.

19.
- **a.** $2\mathbf{u}$
- **b.** $\mathbf{u} + \mathbf{v}$
- **c.** $\mathbf{u} - \mathbf{v}$ (you do not have to draw it as a position vector)
- **d.** $\text{proj}_{\mathbf{v}}\mathbf{u}$

Use a graph to perform the indicated operations (provide labels).

20. Plot the point $(3, 3, 3)$.
- **a.** Sketch the plane $(0, 3, 0)$.

Perform the indicated operation.

21. Let $\mathbf{u} = 5\mathbf{i} + 4\mathbf{j}$ and $\mathbf{v} = 3\mathbf{i} - \mathbf{j}$. Find the angle between $\mathbf{u}$ and $\mathbf{v}$.

22. Find a vector perpendicular to $\mathbf{u} = \langle 2, 0, 0\rangle$ and $\mathbf{v} = \langle 0, -3, 0\rangle$

23. A bullet is fired with a muzzle velocity of 1037 ft/sec from a gun aimed at an angle of 8 above the horizontal. Find a vector describing the vertical and horizontal components of the bullet's velocity.

Derivatives of multi-variable functions

Find the indicated partial derivative of the following functions.

24. Find g_y and g_y for $g(x, y) = (x^2 + 2)e^y$.

25. Find the four second partial derivatives of $h(x, y) = x^2 \cos 2y$.

Use the Chain Rule to find the following derivative. When feasible express your answer in terms of the independent variable.

26. dz/dt where $z = x^3y - x$, $x = t^2$, and $y = 2t^{-1}$.

Maximum/Minimum of multi-variable functions

Compute the gradient of the following function and evaluate it at the given point P.

27. $f(x, y) = x^2 + y^2$; $P(-1, 2)$

Consider the following functions and points P.

a. *Find the <u>unit</u> vectors that give the direction of steepest ascent and steepest descent at P.*

b. *Find a vector that points in a direction of no change in the function at P.*

28. $f(x, y) = x^2 + y^2$; $P(-1, 2)$

Use the Second Derivative Test to determine (if possible) whether each critical point of the given function is a local maximum, local minimum, or saddle point.

29. $f(x, y) = 4 + x^3 + y^3 - 3xy$ with critical points $(0, 0)$ and $(1, 1)$.

Find all of the critical points (there are two) of the following function.

30. $f(x, y) = \dfrac{x^3}{3} - \dfrac{y^3}{3} + 3xy$

Lagrange Multipliers

Use Lagrange multipliers to find the maximum and minimum values of f (when they exist) subject to the constraint.

31. $f(x, y) = x + 2y$ subject to $x^2 + y^2 = 4$
32. $f(x, y) = e^{2xy}$ subject to $x^3 + y^3 = 16$